

Mobile Testing

Android, iOS, real devices, emulators, Appium and device-cloud platforms.

What changes in Mobile Testing

Mobile applications deal with different operating systems, screen sizes, hardware, permissions, networks and interruptions. Testing only the functional flow is not enough.

Native, hybrid, WebView and PWA

Native apps use platform-specific technologies. Hybrid apps combine web technologies with a native container. WebView embeds web content. A PWA is a web application with progressive capabilities.

Real device vs emulator/simulator

Emulators and simulators are useful for speed and initial coverage. Real devices are important for hardware, performance, camera, biometrics, notifications and real network behavior.

Essential scenarios

Installation, upgrade, login, permissions, rotation, background/foreground, calls, battery, GPS, camera, keyboard, offline/online, slow networks and interruptions should be considered according to the product.

Appium

Appium is a mobile automation framework that drives applications through platform automation APIs. It can be combined with languages such as Python, Java and JavaScript.

BrowserStack and device clouds

Device-cloud platforms provide remote access to browsers and real devices. They do not replace Appium; they can host automated and manual executions across many combinations.

Android and iOS

Android uses artifacts such as APK/AAB and tools such as ADB/Android Studio. iOS uses IPA and the Xcode ecosystem. Restrictions, permissions and distribution differ.

Mobile strategy

Prioritize device combinations using user data, analytics, risk and official support. Infinite coverage is impossible; build a defensible test matrix.

QA tip

Combine emulators/simulators with a sample of real devices for risks that depend on hardware and operating system.

Official resources

Site oficial: <https://appium.io/>